Digital Readiness and TPACK Skills of ESL Teachers in Tamil Nadu's Semi-Urban Districts: Issues and Implications

M. Nagalakshmi and A. A. Jayashree Prabhakar

Abstract

This study assesses the TPACK (Technological Pedagogical Content Knowledge) abilities of ESL (English as a Second Language) teachers in the Kanchipuram and Chengalpattu districts of Tamil Nadu. It examines the challenges and opportunities of using digital English Language Teaching (ELT). Digital literacy has emphasized the urgent necessity for language teachers to use technology effectively in pedagogical and content areas. Nevertheless, in semi-urban Indian settings, where training and resource situations are typically constrained, this merging becomes challenging. Using a mixed-methods design, the research collected data through a standardized survey (n = 80) and semi-structured interviews (n = 15) with ESL teachers from colleges in Kanchipuram and Chengalpattu districts of Tamil Nadu. The quantitative instrument was based on the validated TPACK framework [1], and qualitative data explored personal experiences and institutional support systems. Data were analyzed using SPSS and WarpPLS to identify significant relationships among TPACK domains, perceived challenges, and digital ELT readiness. Instrument reliability was established using Cronbach's Alpha (α = .72–.88), and construct validity was evidenced using KMO (.82) and Bartlett's Test ($p < .001$). The results revealed that participants exhibited moderate TPACK competency levels, and Technological Knowledge (TK) was identified as the most underdeveloped of these domains. The most significant challenges faced by them were insufficient infrastructure, limited exposure to ICT training, and inadequate support from the administration, despite a large number of faculty members showing keen interest and a positive attitude towards digital training tools in ESL teaching. They take this as a challenge to develop their skills and meet the standards of urban teachers.

Keywords

Content knowledge · Digital literacy · English as a second language · Online teaching · Semi-Urban districts · Tamil Nadu · Teacher awareness · TPACK

1 Introduction

1.1 Background

Online teaching has accelerated after COVID-19. It gives a newer dimension to the teaching–learning process. This paradigm shift is necessary for all teachers to equip themselves to teach today's students. Blended learning was mandatory, and faculty were forced to adapt to the situation. Technology is an essential tool for all faculty members, and they must enhance their digital competencies to meet the needs of students (Trust et al. 2017). The TPACK framework (Mishra and Koehler 2006) emphasizes that faculty should equip themselves with digital tools to match the updated curricula and modern pedagogical strategies, thereby teaching learning objectives and achieving the outlined course outcomes.

1.2 Gap

Research on digital teaching methods in technologically advanced and urban institutions is expanding; however, few studies examine the digital competencies of ESL teachers in India's semi-urban areas. Semi-urban teachers face challenges due to poor infrastructure, limited access to professional development, and a lack of support from policy

M. Nagalakshmi (✉) · A. A. Jayashree Prabhakar
Professor, Department of English, Vels Institute of Science, Technology and Advanced Studies, Chennai, Tamil Nadu, India
e-mail: nagalakshmi.sl@vistas.ac.in

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

A. J. Obaid et al. (eds.), *Emerging Technologies for Next Generation Learning and Education Spaces*, Advances in Science, Technology & Innovation, https://doi.org/10.1007/978-3-032-21822-3_37

levels, all of which hinder the effective integration of digital tools into English language teaching. While national programs like SWAYAM and DIKSHA try to implement digital education, contextual evidence about the readiness of ESL faculty in semi-urban Tamil Nadu is limited.

1.3 Aim

The current research aims to evaluate the TPACK skills of ESL educators in Tamil Nadu's semi-urban areas, identify the challenges they face in using digital tools to teach English, and explore potential future directions for applying digital pedagogy in these learning environments.

2 Literature Review

2.1 Synopsis of the TPACK Framework

Drawing on (Shulman 1986) Pedagogical Content Knowledge (PCK), the TPACK (Technological Pedagogical Content Knowledge) model created by Mishra and Koehler (Mishra and Koehler 2006) encompasses five distinct knowledge bases: Content, Pedagogy, Technology, their intersections, and the Induction space in which they meet. There are some collaborative and engaging combinations available, which have led to a framework called Technological Pedagogical Content Knowledge (TPACK). The central claim of this TPACK is to build a good rapport with students, enhance their communication skills, and help them integrate this knowledge into other subject teaching. Currently, this model is a significant one for understanding and assessing faculty members' digital knowledge. It helps to comprehend the teachers' technical integration in their teaching and their proficiency with those tools, according to (Koehler et al. 2013). This approach makes sure that online technical tools in English Language Teaching go hand in hand with language learning strategies and linguistic content.

2.2 Online Teaching and Learning Post-COVID-19

This scenario has become the new normal everywhere. Even in our Indian classrooms, the advent of technology and technical tools plays a vital role in teaching and learning. This phenomenon is not limited to urban areas, but also occurs in semi-urban and rural areas, where it quickly adapts to inevitable situations (Huang et al. 2020). As the teaching profession has advanced to the next level of digital competency, models such as TPACK are inevitable for assessing in-service teacher training and policy-making (Jena 2020; Trust et al. 2018).

2.3 Acquiring Language Using TPACK

Technology is a boon to teaching and learning. Especially while learning a language, it can support multimodal teaching, enhance interactive sessions, and provide one-on-one feedback, among other benefits. The TPACK model is highly popular in recent research on language acquisition. For instance, Baren et al. identified that ESL instructors who have received specialized TPACK training were proficient and utilized digital resources to develop students' vocabulary skills, academic writing skills, and integrated work. In Asian ELT settings, there is a lack of context in-service training programs and inadequate access to infrastructure. In-service training will be reflected in the integration of TPACK in ELT settings, as stated by (Chai et al., 2013). Their research actually highlighted where infrastructure and institutional hindrances usually impede high instructor motivation. Wang et al. (2014) proposed a four-phase TPACK model for this type of area, which would incorporate TPACK training into teacher training education. The results of their study indicated a positive sign, and the use and adaptation of TPACK align with communicative language teaching principles.

2.4 Struggles While Implementing Technology-Based Instruction in Indian Semi-Urban and Rural Areas

The efforts to integrate technology-based instruction were diluted due to India's socioeconomic and inadequate infrastructure, especially in semi-urban and rural areas. Most of the instructors from these areas have stated several challenges, including poor network connections, a limited number of technical tools such as computers and smartboards, and a lack of awareness about utilizing proper technology-based instruction.

Furthermore, the flexibility required for ELT classroom innovation is hindered by systemic issues such as large class sizes, overly prescribed curricula, and inadequate institutional support. There is also evidence of a digital divide between teachers in urban and non-urban environments, with the former having fewer access points to technology-enabled training (Trust et al. 2018).

2.5 Teacher Readiness and Digital Proficiency

Teacher readiness is crucial when embracing technology. There are two categories of barriers, according to (Ertmer and Ottenbreit-Leftwich 2010): first-order (e.g., training, lack of access) and second-order (e.g., beliefs, attitudes). Even though many ESL teachers may have access to technology, their confidence and belief in its effectiveness often determine how widely it is used. According to (Graham et al. 2011), teachers' use of TPACK is more impacted by their comfort with experimentation than by policy demands. The Indian ESL context was dominated by the grammar-translation method. Raman and Thannimalai's study indicated that experience and self-competence were the indicators of TPACK in Malaysian teachers. The study focused on institutional support and training as significant factors for TPACK growth. It is emphasized and reinforced in this study.

2.6 Personal Development and Institutional Support

Accepting online pedagogy needs institutional support. The institute should understand the TPACK framework and its potential influence on the teaching and learning process. This recognition will enhance peer collaboration, and faculty members can incorporate it into their digital platforms, according to (Trust et al. 2018). Digitally literate teachers are more likely to be produced by educational institutions that encourage experimentation, offer digital resources, and incentivize creativity. Howard et al. (2021) discovered that teachers frequently reported "digital burnout" in nations with strong policy support but weak institutional follow-through, where they were expected to use tools without the necessary infrastructure or training. It is a growing issue in Indian semi-urban colleges, where systemic support is still lacking despite the post-pandemic expectation of digital ELT.

2.7 An Overview of the Gaps in the Research

Few studies have examined the use of TPACK by ESL teachers in semi-urban Indian (Tamil Nadu) contexts, despite the growing interest in digital pedagogy. The majority of Indian TPACK research focuses on urban private schools or STEM education. Few empirical studies have looked at the difficulties and potential of TPACK in tertiary regional language instruction contexts. This study fills that gap by contextualizing ESL teachers in Tamil Nadu's semi-urban areas, emphasizing both individual agency and systemic obstacles to TPACK adoption.

3 Research Objectives

- To evaluate ESL instructors' current TPACK proficiency in Tamil Nadu's semi-urban undergraduate institutions.
- To determine educators' main obstacles when integrating digital tools into ELT.
- To investigate how educators in semi-urban settings view the potential usage of digital ELT.
- To suggest doable tactics for enhancing ESL faculty members' digital readiness and TPACK-based training.

4 Research Questions

- To what extent are ESL teachers in Tamil Nadu's semi-urban areas proficient in the TPACK framework?
- What are the main obstacles to integrating digital technology in ELT from the teachers' viewpoint?
- How do educators view the use of digital ELT in their schools going forward?
- How can institutional support improve TPACK-based teaching methods?

5 Methodology – Research Design

To ensure that survey items were appropriate, clear, and aligned with the intended constructs, a thorough piloting procedure was employed. The survey was piloted with ten ESL instructors and evaluated by five language education specialists. The reliability and contextual relevance of the tool were improved through changes to item wording, sequencing, and Likert scaling based on feedback from this process. This study uses a mixed-methods design that blends quantitative and qualitative techniques to thoroughly grasp ESL teachers' TPACK competencies and their experiences with digital ELT. The qualitative component uses semi-structured interviews to explore contextual challenges and perceptions, while the quantitative component uses a structured survey to assess self-reported TPACK knowledge levels.

5.1 Participants

A group of faculty members from colleges in Kanchipuram and Chengalpattu districts were involved in this study. These

two districts are located in Tamil Nadu's semi-urban areas. The selection is based on their experience in teaching English to UG students. A small number of professors participated in one-on-one interviews to ensure the quality of the survey.

5.2 Research Tools

The primary tool used to collect the required data was a questionnaire, which was designed based on the TPACK framework. The semi-structured interview was also based on the TPACK framework. A five-point Likert scale, ranging from "Strongly Agree" to "Strongly Disagree", was framed to incorporate all seven domains of TPACK (Schmidt et al. 2009). The interview protocol gathered information on teachers' attitudes toward digital pedagogy in ELT, institutional support levels, and lived experiences.

5.3 Justification for Sampling and Ethical Issues

Purposive sampling was employed in the study to select eighty ESL instructors who teach at institutions in Kanchipuram and Chengalpattu, the semi-urban districts of Tamil Nadu. These areas were selected because, despite having limited access to ICT resources at the urban scale, they experienced a rise in online and blended English instruction following the pandemic. Participants had to be actively involved in undergraduate English courses and have taught for at least two years. The research ethics committee of the affiliated university granted ethical approval. In both the quantitative and qualitative stages of the study, each participant gave their informed consent and was guaranteed anonymity and confidentiality.

5.4 Data Analysis

Descriptive statistics were used to analyze the TPACK survey's quantitative data to calculate domain-wise competency levels, mean scores, and standard deviations. Relationships between TPACK domains and external factors, such as institutional support, were investigated using inferential statistics, including regression and correlation analyses. The study's conceptual model was further validated using WarpPLS for Structural Equation Modelling (SEM). In order to identify recurrent patterns, obstacles, and opportunities as perceived by participants, qualitative data from interviews will be transcribed and thematically coded using NVivo. Cronbach's Alpha was used to assess the survey instrument's internal consistency before analysis began, and all domains had α values greater than 0.70, which is an acceptable threshold. Additionally, construct validity was confirmed using the Kaiser–Meyer–Olkin (KMO) measure (0.82) and Bartlett's Test of Sphericity ($\chi 2 = 1456.2$, $p < 0.001$), validating the instrument's suitability for multivariate analysis and factor exploration.

Table 1 TPACK domain, mean and standard deviation (*Source* Authors)

TPACK domain	Mean	Standard deviation
Technological Knowledge (TK)	3.1	0.85
Pedagogical Knowledge (PK)	4.2	0.60
Content Knowledge (CK)	4.3	0.57
TPACK ıntegration domains	3.5	0.70
Challenges and prospects	3.2	0.88

5.5 Study Limitations

Despite being thorough, the study has some limitations. Response bias can be imposed by the TPACK survey's reliance on self-report information, particularly in dimensions related to technological confidence. In addition, the findings are unlikely to be as universally generalizable as they might otherwise be due to purposive sampling of respondents from the Kanchipuram and Chengalpattu districts of Tamil Nadu. To triangulate the data further, subsequent studies could employ a larger, more geographically representative sample and incorporate potential classroom observations or student comments.

6 Instrument Structure

In order to assess the technological, pedagogical, and content knowledge of ESL teachers in semi-urban Tamil Nadu, the survey instrument utilized in this study was designed in accordance with the TPACK framework (Mishra and Koehler 2006). Integrating perceived opportunities through evaluating contextual challenges while using digital techniques and tools in language instruction. All twenty-five items, with respect to a five-point Likert scale, are in each of the five classifications. The collected responses from each respondent were analyzed. Expert confirmation was sought for the devised questionnaire, and a proper pilot study was conducted to ensure the credibility and validity of the tool. The Cronbach's alpha coefficient above 0.8 for each domain indicated the higher reliability in the pilot test (Table 1).

6.1 Technical Session (TK)

This session analyzed the respondents' ease, awareness and confidence in using educational technology. Evaluating proficiency in implementing technology in the classroom is the primary focus of this technical session.

Example items:

- I can pick up new technological skills on my own.
- I feel comfortable incorporating audio and video multimedia into my lessons.

6.2 Practical Session (PK)

Practical knowledge of teaching and learning was evaluated in this part. It also evaluated the lesson plan, assessment plan, teaching and classroom management, and whether the strategies and techniques were student-centred.

Example items:

- To meet the needs of my students, I employ a range of teaching techniques.
- I feel at ease designing educational activities that promote student engagement.

6.3 Conceptual Session (CK)

This session evaluated the instructors' proficiency in English language components, including grammar, academic writing skills, vocabulary, passive and active skills, and their ability to interact.

Example items:

- I am proficient in English writing conventions and grammar.
- I can simplify complicated English language rules.

6.4 Technical, Pedagogical, Conceptual, and TPACK

This session explored the combination of technical, pedagogical, and conceptual expertise that instructors used cohesively and coherently in the classroom.

Example items:

- I can utilize specific pedagogical strategies using technology (TPK) effectively.
- I can select digital resources that are most suitable for specific language content areas (TCK).
- I know how to effectively teach the English language (proficiency) (PCK).

6.5 Challenges and Prospects

This section explored institutional, personal, and infrastructural barriers to technology integration and assessed teachers' future outlook on digital ELT.

Example items:

- My institution offers sufficient support and digital infrastructure.
- I can attend frequent workshops or training sessions on digital tools.
- I have trouble using digital resources or the internet when teaching.

6.6 Testing for Validity and Reliability

For every TPACK model domain, the internal consistency of the survey instrument was verified using Cronbach's Alpha. With the following alpha coefficients, every subscale showed internal reliability ranging from acceptable to excellent (Table 2 and Fig. 1).

These results demonstrate that each domain's instrument items consistently assess the desired constructs. In general, the study reveals that localized interventions through infrastructure investment, continuous teacher education, and organizational incentives are needed to encourage TPACK uptake and sustain the digital revolution in ESL teaching. The Kaiser–Meyer–Olkin (KMO) scale and Bartlett's Test of Sphericity tested the instrument's construct validity. Statistical significance was obtained through Bartlett's Test ($\chi 2 = 1456.2$, df $=$ 300, $p < 0.001$), and favorable sampling adequacy was reflected in KMO $= 0.82$. These results confirm that the TPACK-aligned instrument structure is construct-valid and that the dataset is suitable for exploratory factor analysis.

Table 2 Testing for validity and reliability (Cronbach's alpha) (*Source* Authors)

Domain	Items	Cronbach's alpha (α)
Technological knowledge	TK1–TK5	0.74
Pedagogical knowledge	PK1–PK5	0.86
Content knowledge	CK1–CK5	0.88
TPACK ıntegration	TPACK1–5	0.81
Challenges and prospects	CP1–CP5	0.72

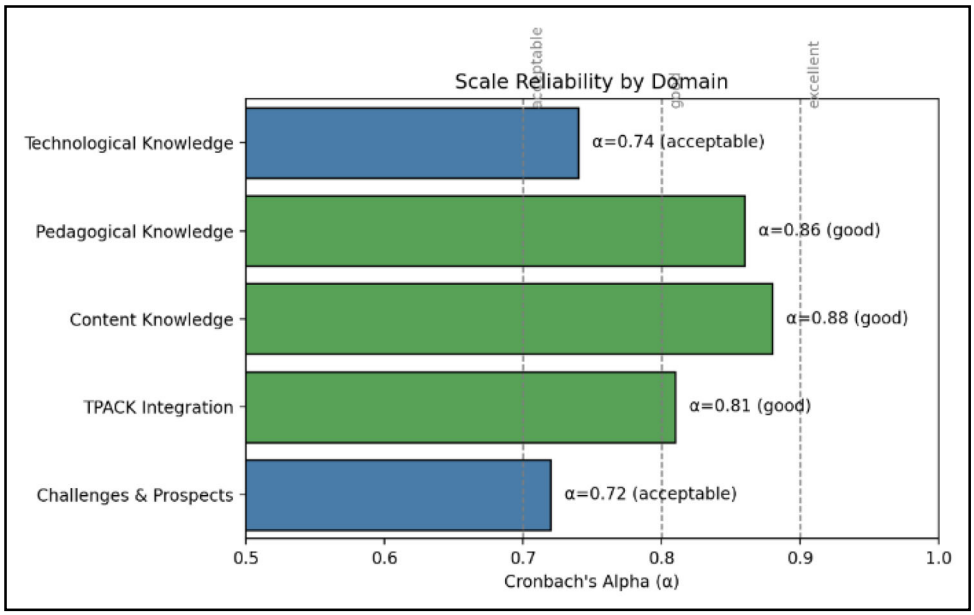


Fig. 1 Cronbach's alpha and scale reliability (*Source* Authors)

7 Results and Discussion

The study's conclusions, derived from the TPACK survey and interview data, are presented in this section. Three themes emerged from the analysis, which matched the goals of the study: TPACK competency levels, ESL teachers' difficulties, and their perspectives on digital ELT.

7.1 TPACK Proficiency

Levels by Discipline. According to the quantitative analysis, ESL teachers demonstrated a strong foundation in English language content and teaching methodologies, displaying moderate to high levels of Pedagogical Knowledge (PK) and Content Knowledge (CK). However, Technological Knowledge (TK) scored significantly lower, indicating that they had limited exposure to or training in educational technologies. The integrated TPACK framework, which needs to synthesize all three knowledge domains, also establishes reasonable competency. Although instructors may possess pedagogical and conceptual knowledge, these results align with other studies demonstrating that they often lack the technological integration skills required for successful digital teaching (Trust et al. 2018). All five sections had their means and standard deviations determined. The outcomes are calculated and given as follows (Fig. 2). The findings suggest that participants are more confident in their knowledge of pedagogy and content than in their ability to use technology effectively or to integrate it with pedagogy and content. These results align with previous studies that highlight disparities in resource availability within teacher training programs, particularly in the semi-urban Kanchipuram and Chengalpattu regions (Scherer et al. 2019; Trust et al. 2018). The following section summarizes these findings and offers valuable suggestions for educational technology and teacher training.

7.2 Data Analysis

The research used quantitative and qualitative data analysis methods. Quantitative data were analyzed using SPSS (version 26) and WarpPLS (version 8.0) software. The analysis followed these steps:

- Reliability Analysis: Cronbach's Alpha was computed for each domain (TK, PK, CK, TPK, TCK, PCK, TPACK) to assess internal consistency ($\alpha > 0.70$ considered acceptable).
- Descriptive Statistics: Means, standard deviations, and frequency distributions were calculated to summarize teacher responses across TPACK domains.
- Inferential Statistics:
  - Pearson Correlation was used to assess relationships between TPACK components.
  - Independent Samples t-tests were conducted to compare mean differences based on demographic variables (e.g., gender, training exposure).
  - One-way ANOVA tested group differences across teaching experience categories.

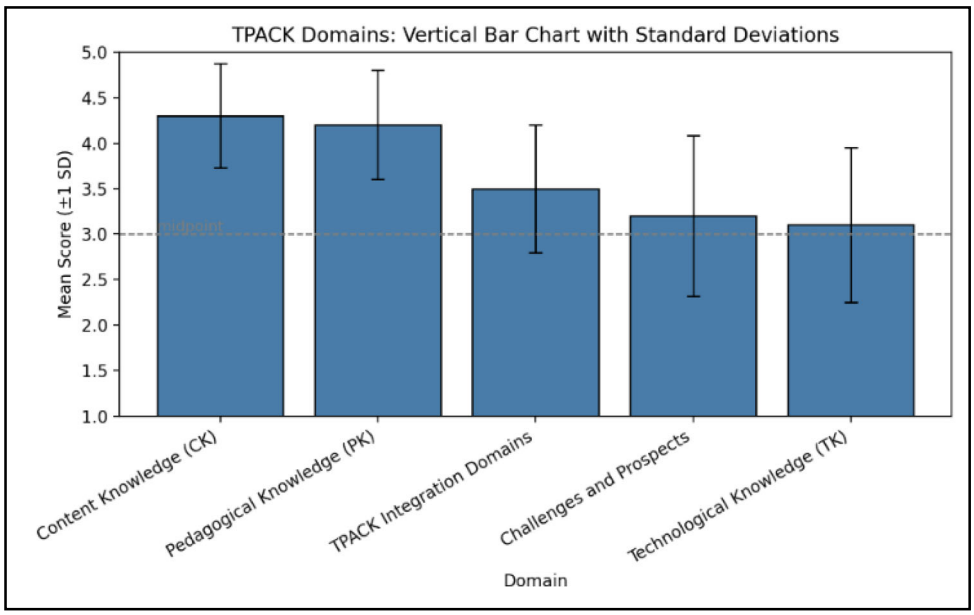


Fig. 2 TPACK domain vertical bar chart (*Source* Authors)

Table 3 Structural equation modelling (*Source* Authors)

Fit ındex	Obtained value	Threshold/acceptable range
Average Path Coefficient (APC)	0.43, $p < 0.001$	Should be significant ($p < 0.05$)
Average Variance Inflation Factor (AVIF)	2.15	≤5.0 (ideal < 3.3)
Tenenhaus GoF (Goodness of Fit)	0.32	Small ≥ 0.1, Medium ≥ 0.25, Large ≥ 0.36

Table 4 Comparative interpretation (*Source* Authors)

Hypothesized path	β (standardized coefficient)	p-value
TK → TPACK	0.29	<0.01
PK → TPACK	0.38	<0.001
CK → TPACK	0.33	<0.001
Institutional support → TPACK	0.27	<0.05
TK × ınstitutional support → TPACK (mediation)	0.18	<0.05

- Multiple Regression Analysis: To determine predictors of overall TPACK competency, a regression model was constructed with TK, PK, CK, and Institutional Support as independent variables.
- Structural Equation Modelling (SEM): Using WarpPLS, the theoretical model was tested for path relationships among TK, PK, CK, and TPACK, with Institutional Support as a mediating variable. Model fit indices (e.g., APC, AVIF, Tenenhaus GoF) were examined for validation. NVivo 12 was used to code the qualitative interview data thematically, and new trends were grouped into themes like perceived prospects, institutional barriers, and technological confidence. Findings from both data streams were validated through triangulation. Tenenhaus GoF = 0.32, Average Variance Inflation Factor (AVIF) = 2.15, and Average Path Coefficient (APC) = 0.43 ($p < 0.001$) were all acceptable fit indices for the SEM model. According to these indicators, the model's explanatory power and reliability are satisfactory (Harris and Hofer 2009) (see Tables 3 and 4).

7.3 Comparative Interpretation of TPACK Domains

The gap in Technological Knowledge points to a systemic pattern observed in many post-colonial educational settings, where digital infrastructure is unevenly distributed, despite the study confirming higher pedagogical and content knowledge levels among participants. Despite having good teaching practices, teachers in semi-urban areas often lack organized access to technology training, which could stifle creativity. In contrast, ICT is frequently incorporated into teacher education programs at urban institutions. These analogies highlight the importance of decentralizing policy to provide digital teacher training in underserved areas. Secondly, instructors who are amenable to utilizing Google Meet or Kahoot demonstrate a positive attitude that can evolve into full digital proficiency with the proper support. The findings indicate that localized interventions involving infrastructure investment, ongoing teacher training, and

organizational incentives are necessary to promote TPACK adoption and sustain the digital revolution in ESL instruction.

8 Findings and Suggestions

The study's instrument showed good psychometric qualities. High sample adequacy (KMO = 0.82) and a significant Bartlett's test ($p < 0.001$), with reliability scores for all domains exceeding the acceptable threshold ($\alpha = 0.72$–0.88), substantiated factor validity. These results enhance the credibility of the findings and confirm the validity of TPACK-based frameworks.

8.1 Findings

This study investigated the TPACK competencies of ESL teachers in Tamil Nadu's semi-urban districts, such as Kanchipuram and Chengalpattu, as well as the difficulties they encounter and the opportunities they see for digital ELT. Despite strong pedagogical and content knowledge, teachers' technological competencies and their ability to integrate these skills into a coherent TPACK framework are relatively underdeveloped, according to the findings. Administrative support problems, limited professional development activities, and infrastructural challenges further impede their ability to adapt and sustain digital pedagogical practices. Most teachers are hopeful and wish to enhance their digital pedagogical competencies in spite of these challenges. This initiative offers hope by aiming to increase capacity, a promising starting point. The findings hold general implications for teacher education institutions, curriculum developers, and policymakers. This model could be extended to other states in subsequent studies, or longitudinal measures could be used to assess the shifts in TPACK over time. Researchers should investigate how teachers' TPACK levels impact students' learning outcomes.

8.2 Suggestions

- Focused TPACK Training Programs: Educational officials should develop and implement workshops for professional development focused on increasing the technological and integrative teaching skills of semi-urban ESL teachers.
- Infrastructure and Resource Allocation: To facilitate hybrid or online models of learning in language teaching, institutions need to give high priority to investments in hardware, internet infrastructure, and accessible digital platforms.
- Peer Mentoring and Collaboration: In order to encourage the sharing of knowledge and to increasingly develop all teachers' digital competence, peer mentoring programs that pair experienced and technically proficient faculty with novice users need to be implemented.
- Policy Support and Administrative Encouragement: Government and institutional entities should incorporate digital pedagogical expectations into policy documents and performance evaluations to reward and recognize teachers for their innovative approaches to ELT.
- Localized Frameworks for Digital ELT: Training materials and digital integration techniques should be tailored to semi-urban areas' (Kanchipuram and Chengalpattu) sociocultural and environmental conditions rather than using one-size-fits-all approaches.

By tackling these issues, institutions and policymakers can close the current gaps in TPACK competencies and enable ESL teachers to run productive, engaging, and technologically enhanced English classrooms in semi-urban India.

8.3 Implications for Policy

According to the findings, systemic changes must be implemented in digital ELT planning. Institutions must align with the National Education Policy (NEP) 2020 to ensure ICT readiness at the tertiary level, particularly in regional and semi-urban colleges. Sustainability for the long term would be ensured by making it mandatory to include TPACK-aligned digital pedagogy modules in B.Ed. and M.A. English programs. State education agencies might also launch incentive programs to reward teachers who use computer-delivered ELT innovatively in low-resourced environments. The pedagogical environment of English language teaching in India would be enriched by such initiatives, which would further enhance digital equity. Integrating these modules within the designed curricula, the TPACK theoretical underpinnings might have been successfully implemented, and the teachers would gain some practical knowledge and experience while taking their training programs. İt would help them to teach real-world classroom contexts. This basic training emphasizes the significance of digital literacy, as well as the necessity of inclusion in today's demanding digital literacy world. They gain competencies in pedagogy and content-relevant application of technology. This would bridge the existing skill gap between pre-service and in-service teachers, as the present study highlighted that many teachers from semi-urban areas rely heavily on self-directed learning for digital skills. The NEP 2020 places significant emphasis on digital readiness and empowerment of teachers.

References

Chai, C. S., Koh, J. H. L., & Tsai, C.-C. (2013). A review of technological pedagogical content knowledge. *Journal of Educational Technology & Society*, *16*(2), 31-51.

Ertmer, P. A., & Ottenbreit-Leftwich, A. T. (2010). Teacher technology change: How knowledge, confidence, beliefs, and culture intersect. *Journal of Research on Technology in Education*, *42*(3), 255-284.

Graham, S., Harris, K., & Hebert, M. A. (2011). *Informing writing: The benefits of formative assessment*.

Harris, J., & Hofer, M. (2009). Instructional planning activity types as vehicles for curriculum-based TPACK development. Society for Information Technology & Teacher Education International Conference, Charleston, SC, USA, pp. 4087–4095.

Howard, S. K., Tondeur, J., Ma, J., & Yang, J. (2021). What to teach? Strategies for developing digital competency in preservice teacher training. *Computers & Education*, *165*, 104149.

Huang, R. H., Liu, D., Tlili, A., Yang, J., Wang, H., & Zhang, M. (2020). *Handbook on facilitating flexible learning during educational disruption: The Chinese experience in maintaining undisrupted learning in COVID-19 outbreak*. Beijing: Smart Learning Institute of Beijing Normal University.

Jena, P. K. (2020). Impact of Covid-19 on higher education in India. *International Journal of Advanced Education and Research (IJAER)*, *5*(3), 77-81.

Koehler, M. J., Mishra, P., Kereluik, K., Shin, T. S., & Graham, C. R. (2013). The technological pedagogical content knowledge framework. In *Handbook of research on educational communications and technology* (pp. 101–111). Springer.

Mishra, P., & Koehler, M. J. (2006). Technological pedagogical content knowledge: A framework for teacher knowledge. *Teachers College Record: The Voice of Scholarship in Education*, *108*(6), 1017-1054.

Scherer, R., Siddiq, F., & Tondeur, J. (2019). The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education. *Computers & Education*, *128*, 13-35.

Schmidt, D. A., Baran, E., Thompson, A. D., Mishra, P., Koehler, M. J., & Shin, T. S. (2009). Technological pedagogical content knowledge (TPACK) the development and validation of an assessment instrument for preservice teachers. *Journal of Research on Technology in Education*, *42*(2), 123-149.

Shulman, L. S. (1986). Paradigms and research programs in the study of teaching: A contemporary perspective. *Handbook of research on teaching*, 3–36.

Trust, T., Carpenter, J. P., & Krutka, D. G. (2017). Moving beyond silos: Professional learning networks in higher education. *The Internet and Higher Education*, *35*, 1-11.

Trust, T., Carpenter, J. P., & Krutka, D. G. (2018). Leading by learning: Exploring the professional learning networks of instructional leaders. *Educational Media International*, *55*(2), 137-152.

Wang, S.-K., Hsu, H.-Y., Campbell, T., Coster, D. C., & Longhurst, M. (2014). An investigation of middle school science teachers and students use of technology inside and outside of classrooms: Considering whether digital natives are more technology savvy than their teachers. *Educational Technology Research and Development*, *62*, 637-662.